EXPLORATORY ANALYSIS OF MOBILE APP USAGE AND RATINGS

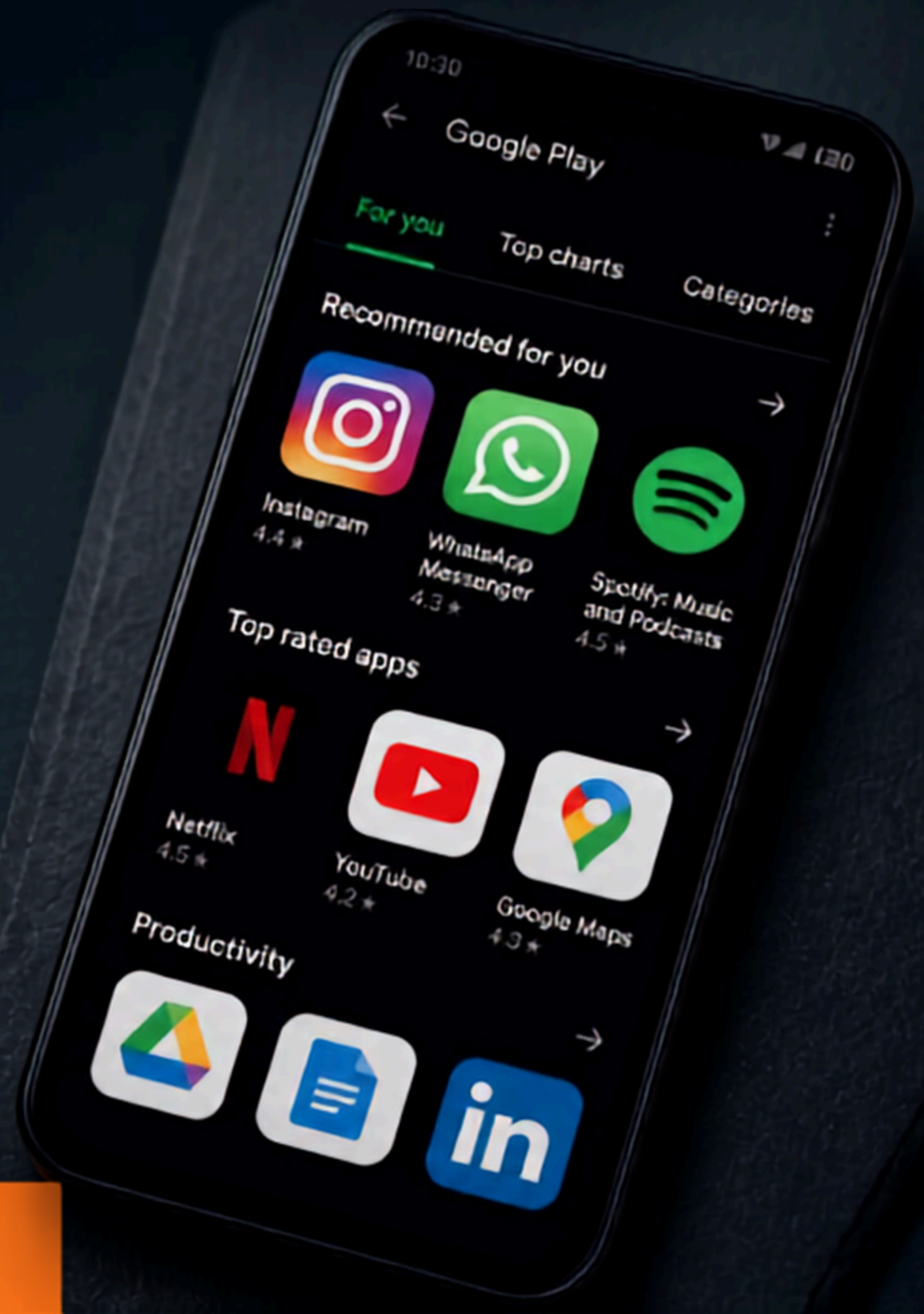
Mini Project – Data Science

Domain: Exploratory Data Analysis (EDA)

Technologies: Python, Pandas, NumPy, Matplotlib, Seaborn

Dataset: Google Play Store App Dataset

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INTRODUCTION

Mobile apps are part of everyday life

App stores offer thousands of applications

Each app varies in ratings, reviews, categories, and installations

Exploratory data analysis uncovers patterns and valuable insights

This project examines mobile app usage and ratings



PROBLEM STATEMENT

Thousands of apps make it hard to find the most popular and highly rated.

User ratings and reviews hold valuable insights.

Popularity and downloads vary across app categories.

We analyze these factors with data visualization to reveal user preferences and app popularity.



OBJECTIVES

Explore the Google Play Store dataset

Prepare and clean the data

Examine categories and ratings

Compare free vs. paid apps

Identify top-rated and most popular apps

Reveal key patterns through visualization



DATASET DESCRIPTION

Google Play Store dataset

Contains information about mobile applications

Includes key attributes such as app, category, rating, reviews, installs, type, and price

Well-suited for exploratory data analysis



TECHNOLOGIES USED

Python as the main programming language
Pandas for data loading and manipulation
NumPy for numerical operations
Matplotlib and Seaborn for visualization
Google Colab / Jupiter Notebook for development

WHY PYTHON?

Easy to learn and use
Backed by powerful data analysis libraries
Ideal for visualization and exploratory analysis



DATA PREPROCESSING

Load and explore the dataset

Detect and handle missing ratings

Remove duplicates and inconsistencies

Standardize and clean data values

Prepare a clean dataset for analysis



RAW DATA → **CLEANING** → **PREPROCESSING** → **CLEAN DATASET**

EXPLORATORY
DATA ANALYSIS & VISUALIZATION

Understand how apps are used and rated

Compare categories and pricing patterns

Identify top apps by reviews and installs

Explore how ratings relate to reviews

Reveal patterns that drive insights



RESULTS & FINDINGS

The majority of apps are free

Ratings cluster between 3.5 and 4.5

Some categories have far more apps than others

More reviews are associated with higher user engagement

Install data helps reveal which apps are most popular

Exploratory analysis uncovers patterns that drive smarter decisions



CONCLUSION & FUTURE SCOPE

Exploratory analysis reveals meaningful patterns in mobile app data

Clean, well-prepared data strengthens the quality and reliability of insights

Visual analysis brings app trends into clear focus

Ratings, reviews, categories, and installations drive valuable insights

Python libraries make the process efficient and effective

